

Technical investigation report

Controlled dataset experiment - annotated examination

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Case identifier	4ab70cc1-3819-417d-a34c-9aae27c9bb73
Revision	v1.1 - revised from the supplied technical export and saved analysis

1. Purpose, scope and evidence identity

The examination question is whether processes in the image contain executable memory that warrants further investigation, and what interpretation is supported by the observed artifacts. Process selection follows the triage evidence; prior knowledge of the sample identity is not used to establish malicious behavior.

This report covers five lead objects and five attention-ranked region exhibits from one image. The original export selected three findings and one region; this revision also incorporates the saved logging/task artifacts and remaining top-five region analyses. Unselected leads are not assessed or cleared.

Image	2580_5.vmem
Recorded size	4,412,228,315 bytes
Recorded SHA-256	777d71d7106e5ded19592c075058da12049bfcd658221e70f0579ad4bbd9cff4

The recorded hash identifies the examined input. A complete original acquisition procedure and transfer/custody history are not reconstructed. This section is evidence identification, not a claim of a complete chain of custody.

Method and coverage

Saved Volatility 3 Framework 2.28.0 outputs, process inventory, command lines, malfind, privileges, network and scheduled-task records were reviewed alongside the completed VADViT analysis. Triage requested 22 plugins and reports no extraction failures. Processing records include reuse of local plugin artifacts. No new acquisition or pipeline execution was performed for this revision.

The aggressive profile surfaced 33 lead objects across an inventory of 118 processes. Scores rank review priority and are not validated probabilities of compromise. For PID 2580, 80 VADs were extracted, 49 grid regions were ranked, and five received saved region analysis. This is a focused examination, not exhaustive reverse engineering.

Evidence times are quoted in UTC. The image reports system time 2024-12-27T07:57:04Z; this is a value from the image, not an independently verified acquisition timestamp. Saved analysis generation and report preparation occurred later.

2. Process prioritization and collection context

The selected candidate and two PowerShell processes each scored 23.6/30. The private executable allocation and loader-related sequence distinguish the next examination question; a filename or equal risk score does not supply the conclusion.

E1. malware.exe - PID 2580

Behavioral interpretation unresolved; loader/payload hypothesis: moderate confidence

Recorded command	"Z:\malware.exe"
Parent / architecture	PPID 3768 / WOW64
Created (UTC)	2024-12-27T07:55:04Z
Memory lead	0x2ea0000; private PAGE_EXECUTE_READWRITE allocation

Private executable memory and a loader-walking sequence make this process a suitable candidate for deeper examination. The Critical score is shared with two PowerShell processes, so the allocation and instruction evidence provide the basis for prioritization. The filename is not treated as proof of maliciousness. Enabled SeDebugPrivilege indicates capability, not demonstrated token manipulation or credential access.

Recorded launch lineage

Schedule service (svchost.exe, PID 1232) -> PowerShell (PID 3768, log_pid.ps1) -> selected process (PID 2580). PowerShell PID 5088 is a separate child of PID 1232 and runs export_logs.ps1.

Attribution basis: the dataset creator confirms that Task Scheduler and the logging mechanisms were used for dataset creation. The matching command lines support attribution of these specific findings to the harness. This is a provenance statement, not an independent validation of the scripts' full contents.

E2. PowerShell PID 3768

Collection artifact | attribution confidence: high

The log_pid.ps1 command line and scheduler-service parent match the collection mechanism confirmed by the dataset creator. This accounts for the recorded launch of PID 2580 without establishing adversary persistence or execution. Collection context does not explain the purpose of the private executable allocation in the child process. The parent's RWX-memory and privilege indicators do not establish LSASS access.

E3. PowerShell PID 5088

Collection artifact | attribution confidence: high

The export_logs.ps1 command line matches the confirmed dataset logging mechanism. This process shares scheduler-service parent PID 1232 with PID 3768; it is a sibling of the selected process's parent. Its Critical triage score does not establish malicious activity or credential dumping.

E4. LOG scheduled task

Collection artifact | attribution confidence: high

The PowerShell action and export_logs.ps1 arguments match the logging process and confirmed collection configuration. The task is retained as collection instrumentation rather than adversary persistence. Its stored last-run fields predate the reported creation time and are unsuitable for reconstructing the sample execution order.

3. Private executable allocation and working hypothesis

R2. 0x2ea0000 - principal memory exhibit

Address range	0x2ea0000-0x2ea1fff
Size / backing	8,192 bytes / private; no backing file
Protection	PAGE_EXECUTE_READWRITE
Entropy	5.2964 bits per byte; a descriptive measurement, not a packing verdict
Extracted-byte SHA-256	a4428fb76a495592f6698f14fd826f15e2c9b34c084d13d6286c6d0ce09255eb

Checkable instruction and string observations

The following excerpt is taken from the saved x86 listing. Addresses are virtual addresses in PID 2580; offsets in the table are measured from region base 0x2ea0000.

Relative offset	Virtual address	Bytes	Saved x86 instruction
0x0b79	0x2ea0b79	64a130000000	mov eax, dword ptr fs:[0x30]
0x0b7f	0x2ea0b7f	8b400c	mov eax, dword ptr [eax + 0xc]
0x0b82	0x2ea0b82	8985d8efffff	mov dword ptr [ebp - 0x1028], eax

An ASCII URL occurs at relative offset 0x170b (virtual address 0x2ea170b). Presented in defanged form:
hxxp://hi.baidu[.]com/aegifjfttrggluze/item/be185dc989cae4f4984aa0df

Analyst interpretation

The private RWX allocation has no backing file and is also identified by malfind. This is consistent with an executable buffer populated at runtime. At virtual address 0x2ea0b79, the x86 listing reads fs:[0x30]; the subsequent read at [eax + 0xc] is consistent with access to PEB loader structures. Together, these observations support a loader/payload-code interpretation at moderate confidence. The methods examine the same image, so their agreement is not independent ground truth. The branch at 0x2ea0035 appears within a traversal-like sequence; the automated decoder-loop label does not establish decryption. The embedded HTTP URL provides a network-correlation lead, but communication and C2 are unconfirmed. Legitimate runtime-generated code remains an alternative pending allocation and execution tracing.

Working hypothesis developed from the observations

After review of the allocation properties, matching malfind address, and PEB/loader-access instructions, the private allocation at 0x2ea0000 appears consistent with loader or payload code operating within PID 2580. These observations support the interpretation at moderate confidence. The confirmed collection mechanism explains the process launch but not the allocation's origin or purpose. Legitimate runtime-generated code remains a testable alternative. The model's Exploit label does not establish a vulnerability or successful exploitation.

Pattern-location correction: the original export mixed virtual addresses and relative offsets. Decoder candidates at 0x2ea0035 and 0x2ea152c correspond to offsets 0x35 and 0x152c; the PEB access at offset 0xb79 corresponds to virtual address 0x2ea0b79. Candidate labels are retained as review questions, not confirmed behavior.

4. Classifier result and attention-ranked context

The saved analysis records a loaded trained checkpoint. Exploit is the highest class score: 0.804653 (80.47%); HackTool is next at 9.33%. These are class scores, not calibrated probabilities that a vulnerability was exploited or the host was compromised.

Rank	Address	Reported backing	Bytes	Relative attention
1	0x72ce0000	dhcpcsvc.dll	94,208	1.000000
2	0x2ea0000	Private; no backing	8,192	0.629570
3	0x72d80000	cryptsp.dll	86,016	0.429805
4	0x72c20000	ncrypt.dll	155,648	0.408780
5	0x72bd0000	ncryptsslp.dll	131,072	0.389522

Attention values are relative ranking/display values. They are not probabilities of maliciousness, do not sum to 100%, and do not establish causal importance. Four of the top five regions report SysWOW64 DLL backing with PAGE_EXECUTE_WRITECOPY protection.

R1. 0x72ce0000 - dhcpcsvc.dll mapping

Highest model attention falls on a file-backed DLL rather than the private allocation. The parsed PE identifies i386, while the saved listing uses x86-64. Instruction-level, loop, and call-count interpretations are therefore deferred pending correct decoding. High attention and PAGE_EXECUTE_WRITECOPY protection do not establish tampering. The reported backing path is not a verified publisher signature; compare executable pages with a trusted matching DLL.

Extracted-byte SHA-256: 75f6c9b36948891d4955caf25da8321b6fc402154002f3a07e09b37d249a2ec4

R3. 0x72d80000 - cryptsp.dll mapping

The file-backed mapping and copy-on-write executable protection do not independently demonstrate injection. The parsed i386 metadata conflicts with the x86-64 listing. Structure analysis covers 65,536 of 86,016 bytes, limiting completeness. No malicious modification is established; correct decoding and comparison with the matching trusted DLL are required.

Extracted-byte SHA-256: c3c6ce71b479e985f3230c87b830a5837986f6b204d8dea0dd25ed9ba0dd19ef

R4. 0x72c20000 - ncrypt.dll mapping

Decoder-loop and indirect-call findings remain provisional because the x86-64 listing conflicts with i386 PE metadata. Structure analysis covers 65,536 of 155,648 bytes and the instruction listing is capped. Library strings and PEB-access signatures alone do not establish malicious cryptographic activity. Validate the architecture and mapped-page integrity before attributing the region to tampering.

Extracted-byte SHA-256: 71bfbe0ad694821754799c5a0d50e9b09475bddee6c106ba77ee8a0a81dc93cb

R5. 0x72bd0000 - ncryptsslp.dll mapping

No MZ signature was found at the extracted region start. Image layout and byte availability need checking; deliberate header removal is not established. Structure analysis covers 65,536 of 131,072 bytes. Given the WOW64 process and reported SysWOW64 backing, the x86-64 listing requires architecture validation. Cryptography-provider registry strings provide module context, not evidence of registry modification or persistence.

Extracted-byte SHA-256: 4ad168714fb36ebf3f7d72d8c57603de9d1c94a40020edbaa6bfffdf0307c741

Model provenance: Multi_32_224_6f_3u.pt; SHA-256 740ab8f1ad128a93e686739023519960a6394ca7b90d8af07d2ebd3eaf2aec92. Output names use the nine-class labels.json mapping. Checkpoint and mapping availability do not validate generalization or accuracy on this image.

5. Network lead and review of heuristic mappings

E5. Unattributed TCP connection

Ownership / C2 assessment: insufficient evidence

Local endpoint	192.168.124.219:49691
Remote endpoint	172.172.255.217:443
Recorded state / owner	ESTABLISHED / no owning PID

The ESTABLISHED connection has no owning PID and cannot be associated with PID 2580 or the embedded URL. A TCP destination port of 443 does not itself identify the application protocol. No protocol/port mismatch, C2 exchange, or exfiltration is established. The automatic T1571 association is unsupported by this observation and is excluded from the assessment.

ATT&CK labels assessed against the available evidence

The original automatic labels are reviewed below rather than presented as detected techniques or a chronological attack narrative. Collection instrumentation is excluded from adversary attribution. No technique in this table is established as executed adversary behavior.

Automatic association	Assessment in this examination
T1055 / T1055.001 Process / DLL injection	Private executable memory is a lead; neither cross-process injection nor DLL injection is demonstrated. No allocating actor or injection mechanism is established.
T1140 Deobfuscation / decoding	Backward-branch candidates require code review. The traversal-like sequence at 0x2ea0035 does not establish a decoder.
T1106 Native API	PEB/loader access supports a runtime-resolution interpretation. Specific API use is not independently established by that access alone.
T1071.001 Web protocols	An HTTP URL is embedded in memory. Actual web communication and a C2 role are unconfirmed.
T1003.001 LSASS memory	RWX memory and enabled SeDebugPrivilege do not demonstrate LSASS access or dumping. The automatic credential-access correlation is not adopted as a finding.
T1134 Access token manipulation	Enabled privilege is an observed capability; manipulation, impersonation, and privilege escalation are not demonstrated.
T1571 Non-standard port	No application protocol or protocol/port mismatch is established. Missing PID attribution is not evidence of this technique.

Count correction: the original technique table aggregated rule contributions under an "Objects" heading and included broader triage leads. Such counts are not reported here as unique selected objects. The report scope is the five lead objects and five region exhibits explicitly identified in this document.

6. Assessment, follow-up and limitations

The principal finding is an 8 KiB private executable allocation with an observable x86 PEB/loader-access sequence. Its properties support a loader/payload interpretation at moderate confidence; its allocating actor and execution behavior remain unresolved. The trained classifier's Exploit score is a prioritization aid. Successful exploitation, cross-process injection, credential theft, C2, and exfiltration are not established by this review. Scheduler and logging artifacts are accounted for as collection instrumentation, and highly attended DLL mappings require further validation.

Prioritized follow-up

1. Preserve and document

Retain the memory image, recorded hashes, saved plugin outputs, region extracts, checkpoint/label versions, and collection scripts. Document the original acquisition and transfers if a custody history is required.

2. Validate the private code

Reverse engineer 0x2ea0000 and trace allocation and execution in an isolated lab. Distinguish sample-generated loader/payload code, externally introduced code, and legitimate runtime behavior before assigning a mechanism.

3. Correct and compare DLL mappings

Decode the i386/SysWOW64 mappings in the correct architecture. Compare relevant executable pages with trusted DLLs matching the host build, accounting for relocations, imports, and normal copy-on-write behavior.

4. Resolve network ownership

Correlate the embedded URL and unattributed connection with process-specific network telemetry, DNS/proxy records, and packet capture. Establish ownership and actual traffic before making a C2 claim.

5. Test specific allegations

Require direct LSASS-access or dumping evidence before alleging credential theft. Identify and validate an exploit mechanism before naming a vulnerability or claiming successful exploitation.

Material limitations

The review is limited to one image and saved outputs. No disk analysis, packet capture, execution trace, or independent exploit verification was performed. Five of 49 ranked grid regions were reviewed; the 80 extracted VADs are not all represented in that ranking. Three parsed i386 DLLs were listed in x86-64 mode, and several analyses were capped. Reported paths, timestamps, and metadata originate in the image and are not independently authenticated.

The engine combines configured rule confidences, including derived correlations; those figures are not calibrated analyst certainty. Enabled privileges, executable memory, URLs, and ATT&CK mappings remain observations or leads unless behavior is established. A low score or absence of a reviewed finding does not clear the remaining image.

Evidence and reference record

Primary records: supplied Technical report.pdf; saved result.json and lowlevel.json for investigation 4ab70cc1-3819-417d-a34c-9aae27c9bb73; process analysis 65f1fda-d96a-412a-9f13-6404576f899c; saved pslist, cmdline and scheduled_tasks rows; dataset creator's collection-mechanism confirmation. The companion note file contains the extracted-byte hashes for all five region exhibits.

Original PDF SHA-256: 373eefc67c8c66d901ae434e89c655847e8305d0ba070ea34fb083e2bbf69997

Reporting principles: NIST SP 800-86, sections 3.3-3.4 (analysis, alternative explanations and reporting). Technique terminology: MITRE ATT&CK T1571, T1003.001 and T1134. References establish definitions, not case attribution.

<https://nvlpubs.nist.gov/nistpubs/legacy/sp/nistspecialpublication800-86.pdf>

<https://attack.mitre.org/techniques/T1571/>

<https://attack.mitre.org/techniques/T1003/001/>

<https://attack.mitre.org/techniques/T1134/>

This annotated revision corrects presentation and interpretation of existing records. It does not change the memory image, rerun analysis, or replace the saved platform record.